ECON 381: Midterm 2 — Solution Key

Fall 2026

1. Multiple Choice Answers

Question	1	2	3	4	5	6	7	8	9	10
Answer	a	c	d	a	b	d	b	c	a	b

1. **(a)** A1 says the conditional mean of the regression error is zero at every education level. It allows individual errors and does not require constant error spread.

2. **(c)** Random allocation makes training independent of other earnings determinants. Selection based on low previous income reflecting unobserved earning barriers, motivation, or expected unobserved earnings changes does not provide that guarantee.

3. **(d)** Under the null, the two-sided p-value is the probability of a statistic at least as far from zero as the observed statistic. It is not the probability the null is true.

4. **(a)** Under homoskedasticity, $\text{Var}(\hat{\beta}_1) \approx \text{Var}(u)/[n\,\text{Var}(X)]$. Quadrupling n divides variance by 4 and sampling SD by 2.

5. **(b)** For the nonzero null, $t = (61.5 - 55)/7.4 = 0.8784$. Its absolute value is below 1.96, so do not reject at 5%.

6. **(d)** A Type-I error rejects a true null.

7. **(b)** With a common SE and normal reference, $|t| > 1.96$, $p < 0.05$, and a 95% CI excluding zero are equivalent.

8. **(c)** With n fixed, $\text{Var}(\hat{\beta}_1) \approx \text{Var}(u)/[n\,\text{Var}(X)]$. Smaller error variance and larger education variance both lower slope variance and improve precision.

9. **(a)** $\beta_0 = 668$ and $\beta_1 = 659 - 668 = -9$: the group coded one minus the group coded zero. The descriptive gap alone is not a causal effect.

10. **(b)** The same observations and model give the same OLS coefficients. Changing the uncertainty calculation can change their SEs; robust SEs remain valid under homoskedasticity given the other conditions.

2. True or False Answers

(a) **False.** The interval with 1.96 relies on a large-sample normal approximation. A1–A3 and regressor variation do not guarantee accurate 95% coverage with $n = 10$. Coverage describes repeated samples, not a probability that a fixed slope moves within a realized interval.

(b) **False.** The sample residual identities follow from OLS with an intercept. They concern distances from the estimated sample line and do not verify $E[u_i \mid X_i] = 0$ for unobserved errors from the proposed population or causal line.

(c) **True.** The coefficient SE uses one sample to estimate the sampling SD of slopes across repeated samples of the same size. The population sampling SD is fixed for a specified n and population; the estimated SE generally varies with the sample.

(d) **False.** Heteroskedasticity means changing *conditional variance*, $\text{Var}(u_i \mid \text{Education}_i)$, not changing conditional means. It can coexist with A1, $E[u_i \mid \text{Education}_i] = 0$.

3. Analytical Solutions

Data and conventions. The supplied `WAGE2.DTA` has 526 complete wage–education observations. Its variable label identifies `wage` as average hourly earnings; `educ` is years of education. These calculations use the displayed two-decimal coefficients and robust SEs; final numerical answers are rounded to two decimal places.

(a) **Sample equation, standard errors, and n.**

$$\widehat{\text{Wage}}_i = \underset{(0.73)}{-0.90} + \underset{(0.06)}{0.54}\,\text{Education}_i, \qquad n = 526.$$

The parentheses are the robust SEs of the corresponding coefficients, not regression-error SDs.

(b) **Slope and graduate predictions.** One additional year of education is associated with predicted hourly earnings higher by approximately \$0.54 per hour. This is a dollar association, not a percentage return or an established causal effect.

$$\begin{aligned}
\widehat{\text{Wage}}(12) &= -0.90 + 0.54(12) = \boxed{\$5.58/\text{hour}}, \\
\widehat{\text{Wage}}(16) &= -0.90 + 0.54(16) = \boxed{\$7.74/\text{hour}}, \\
\widehat{\text{Wage}}(16) - \widehat{\text{Wage}}(12) &= 0.54(4) = \boxed{\$2.16/\text{hour}}.
\end{aligned}$$

The predictions and difference above use the rounded coefficients shown on the exam.

(c) **Intercept and plausibility.** The fitted hourly wage at zero years of education is about $-\$0.90$. A negative hourly earnings prediction is economically implausible for this positive-earnings outcome and shows a limitation of extending the fitted straight line to this schooling level. Zero is within the observed 0–18-year education range, so it is *not* correct to claim that zero lies outside the sample range. The fitted line need not represent the zero-schooling conditional average well.

(d) **Coefficient SEs and nonzero-null statistic.** The intercept SE, 0.73 dollars per hour, estimates the sampling SD of intercept estimates across repeated samples of 526 workers. The slope SE, 0.06 dollars per hour per year of education, estimates the sampling SD of education-slope estimates. Neither measures wage dispersion among individual workers.

$$t = \frac{\hat{\beta}_1 - 0.40}{SE(\hat{\beta}_1)} = \frac{0.54 - 0.40}{0.06} \approx \boxed{2.33}.$$

The requested null is 0.40. The table's t-statistic 8.84 tests zero using Stata's unrounded estimates, rather than the requested null hypothesis at 0.40.

(e) **Nonzero-null test and p-value range.**

$$H_0 : \beta_1 = 0.40, \qquad H_1 : \beta_1 \neq 0.40.$$

Because $|t| = 2.33 > 1.96$, **reject at 5%**. The statistic is also above 1.65 but below 2.58, so the two-sided normal p-value satisfies

$$\boxed{0.01 < p < 0.05}.$$

The normal approximation gives $2\Phi(-2.33) \approx 0.02$. The Stata t and p columns test $\beta_1 = 0$ and cannot be directly used for $\beta_1 = 0.40$. The displayed `0.00` is rounding of a small positive probability, not an exact zero.

(f) **Intervals and zero-slope decisions at three levels.**

$$\begin{aligned} CI_{90\%} &= 0.54 \pm 1.65(0.06) \\ &\approx \boxed{[0.44, 0.64]}, \\ CI_{99\%} &= 0.54 \pm 2.58(0.06) \\ &\approx \boxed{[0.39, 0.69]}. \end{aligned}$$

The units are dollars per hour per year of education. Both intervals center on 0.54, and the 99% interval is wider. For the 5% decision, use the normal-approximation interval

$$CI_{95\%} = [0.42, 0.66].$$

Stata uses unrounded estimates to compute its reported statistics. Both the reported interval and the interval computed from the displayed coefficients and SEs exclude zero.

Two-sided null	10% level	5% level	1% level
$\beta_1 = 0$	Reject	Reject	Reject

For the zero-slope null, $|t| = 0.54/0.06 = 9.00$ exceeds all three cutoffs (1.65, 1.96, and 2.58), so reject at the 10%, 5%, and 1% significance levels. Equivalently, zero lies outside the 90%, 95%, and 99% intervals. Decreasing the significance level requires stronger evidence; increasing the confidence level widens the interval.

(g) **Assumptions and limits.**

(i) **A1.** $E[u_i \mid \text{Education}_i = x] = 0$ at every education level. For a causal education effect, omitted wage determinants must not vary systematically with schooling in their contribution to wages. Ability or family background can affect both wages and schooling, making A1 doubtful for this causal model. Zero residual sample means are consequences of OLS methods and do not establish this condition for unobserved population errors.

(ii) **A2.** Wage–education pairs are independent and identically distributed. Independent random sampling from a common target population supports it. Workers sharing employers, local labor-market shocks, or sampling clusters can threaten independence. Different sampling populations or selection methods can threaten identical distribution. The coefficient table does not reveal the sampling process and cannot establish A2; $n = 526$ alone is insufficient.

(iii) **A3.** Unusually large outliers are unlikely. Inspect wage and education distributions, scatterplots, unusually high earnings, units, and coding. A small number of extreme observations may dominate a slope. The table alone cannot establish the population condition; the observed range does not prove that all relevant population tails are well behaved.

(iv) **Robust SEs.** The option does not establish homoskedasticity; it permits valid large-sample uncertainty estimates under either homoskedasticity or heteroskedasticity, given the other assumptions. It does not repair a failure of A1 or dependence between workers. A precise, significant coefficient is not by itself a causal education effect.

4. Bonus Solutions

The model and estimation sample are identical in both commands. Adding `robust` changes coefficient-uncertainty estimation, not OLS minimization.

Item	Classification and explanation
(a) OLS coefficients	**Unchanged.** The same data and least-squares objective produce the same fitted coefficients.
(b) R^2	**Unchanged.** Fitted values, residuals, residual sum of squares, and total sum of squares stay the same.
(c) SER / Root MSE	**Unchanged.** $SER = \sqrt{\sum_i \hat{u}_i^2/(n-2)}$ uses the same residuals and sample size. This is distinct from a coefficient SE.
(d) Slope t-statistic	**May change.** Its numerator is unchanged, but its coefficient-SE denominator can change.
(e) Slope 95% CI	**May change.** The estimate stays the same, but a different coefficient SE can change the margin and endpoints.